Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/02/2004
PI Name:	Marr, David W.M. Ph.D.		
Project Title:	A Novel Colloidal Microfluidics Platform for Spaceborne Micro Total Analysis Systems		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80401-1887	**Congressional District:**	7
Comments:	https://		
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	08/01/2001	**End Date:**	01/09/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG9-1364		
Performance Goal No.:			
Performance Goal Text:			

We use novel fabrication and actuation approaches for the creation of colloid-based microfluidic devices, including pumps, valves, and separators. To use novel fabrication and actuation approaches for the creation of colloid-based microfluidic devices. The “lab-on-a-chip” concept has gained significant research interest in recent years because the drastic reduction in length scales will allow tasks to be performed more rapidly, efficiently, and on smaller sample volumes than ever before. Functional systems fabricated to exploit microscale phenomena possess great promise to significantly streamline processes for fundamental research and medical applications in areas such as bioanalysis, medical diagnostics and therapeutics. Such developments will enable a large-scale shift from centralized laboratories to remote point-of-care and benchtop diagnostic facilities.
Seamlessly integrating individual devices capable of single operations is required however to finally bring to fruition the promise of µTAS (micro total analysis systems) as portable laboratories, chemical production facilities, remediation units, and health monitors. As human exploration and development of space extends from low Earth orbit and the

Task Description:	International Space Station to farther reaches of our solar system, both the need for sophisticated diagnostic equipment and the size and weight constraints placed on payload will surely increase. µTAS which are capable of both chemical analysis and isolating, sustaining and examining individual cells will be needed to meet these requirements. Generally, the utility, speed and performance of microfluidic chips increases as the overall device size decreases, particularly for devices that are ultimately designed for human implantation. The need to mix, administer and separate fluids at these length scales has long been a limiting factor in such devices. While some current microfluid handling devices and techniques enable functional devices at microscales, they may also impose significant constraints upon potential device capability, flexibility and performance. Our approach is to develop a microfluidics toolbox based on colloidal particles that will provide a small-scale integrated approach to microfluidics and the µTAS. Seamlessly integrating individual devices capable of single operations is required however to finally bring to fruition the promise of microTAS (micro total analysis systems) as portable laboratories, chemical production facilities, remediation units, and health monitors. As human exploration and development of space extends from low Earth orbit and the International Space Station to farther reaches of our solar system, both the need for sophisticated diagnostic equipment and the size and weight constraints placed on payload will surely increase. microTAS which are capable of both chemical analysis and isolating, sustaining and examining individual cells will be needed to meet these requirements. Generally, the utility, speed and performance of microfluidic chips increases as the overall device size decreases, particularly for devices that are ultimately designed for human implantation. The need to mix, administer and separate fluids at these length scales has long been a limiting factor in such devices. While some current microfluid handling devices and techniques enable functional devices at microscales, they may also impose significant constraints upon potential device capability, flexibility and performance. Our approach is to develop a microfluidics toolbox based on colloidal particles that will provide a small-scale integrated approach to microfluidics and the microTAS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	(Period covering 7/02-6/03) To date, we have made significant progress in our efforts to incorporate colloids as the functional units of microfluidic devices, having created microfluidic pumps, valves, and single particle/cell separators. As discussed in the previous progress report, these devices are suited to operation post priming, that is after network filling. To complement these devices therefore, we have been working on the incorporation of capillary-based pumps within our current designs. In this, flow through microfluidic devices can be driven using the capillary forces at the walls of the channels. Capillary pumps have many advantages; they are self-driven, require little additional processing, and can be designed to generate relatively constant and well-controlled flow rates. These will aid not only the fabrication of the devices we propose, but also can be used to control the relative flow rates of differing streams during analysis. In our studies this year we have demonstrated not only that microscale capillary flows can be readily redirected (T. Vestad, D.W.M. Marr, J. Oakey, Journal of Micromechanics and Microengineering, 14, 1503, 2004) but also that one can take advantage of our techniques for simple microfluidics based computation (T. Vestad, T. Munakata, D.W.M. Marr, Applied Physics Letters, 84, 5074, 2004). As further development of our colloid-based devices we have been incorporating other light-sources that are significantly smaller than our initial designs. These incorporate diode bars only 1 x 100 microns in size as the light source and may therefore be readily integrated within space-based microfluidic devices (R.W. Applegate Jr., J. Squier, T. Vestad, J. Oakey, D.W.M. Marr, Optics Express, 12, 4390, 2004). Finally, colloid sensors based on dielectrophoresis have been studied as the basis of microfluidic flow sensors.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Articles in Peer-reviewed Journals	Vestad T, Munakata T, Marr DWM. "Flow resistance for microfluidic logic operations." Applied Physics Letters. 2004 Jun 21;84(25):5074. https://doi.org/10.1063/1.1764592 , Jun-2004
Articles in Peer-reviewed Journals	Vestad T, Marr DWM, Oakey J. "Flow control for capillary-pumped microfluidic systems." Journal of Micromechanics and Microengineering. 2004 Aug;14(11):1503-6. https://doi.org/10.1088/0960-1317/14/11/010 , Aug-2004
Articles in Peer-reviewed Journals	Applegate RW Jr, Squier J, Vestad T, Oakey J, Marr DWM. "Optical trapping, manipulation, and sorting of cells and colloids in microfluidic systems with diode laser bars." Opt Express. 2004 Sep 20;12(19):4390-8. PubMed PMID: 19483988 ; https://doi.org/10.1364/OPEX.12.004390 , Sep-2004